

SERVICE RUNBOOK

Auth Service

service: auth-service

TIER 1

ON-CALL

1 Service Overview

`auth-service` handles login, session management, token issuance, and signature validation. It maintains a **signing keyset** used to validate both current and legacy tokens, and an **in-memory session cache** for active sessions.

IMPACT This is a **tier-1 service** — an outage locks users out of every dependent product. Treat deploys and key rotations as elevated-risk windows.

2 Common Failure Patterns

Memory growth from the in-memory session cache is a known risk if a deploy introduces changes to cache eviction logic — watch for `OOMKilled` events with steadily climbing memory rather than a sudden spike.

Token validation failures for a subset of users are often linked to **signing key rotation** removing an older key from the validation keyset before all legacy tokens have expired naturally.

Deploys and key rotations should always be treated as **higher-risk windows** for this service, given its history of maintenance windows overrunning their expected duration.

SYMPTOM	LIKELY CAUSE / FIRST CHECK
Pods restarting every 2–3 hrs, <code>OOMKilled</code>	Session cache eviction bug — check for missing TTL policy after recent deploy
<code>session</code> expired immediately post-login	Recent deploy not fully rolled back in all regions — check per-region version

SYMPTOM	LIKELY CAUSE / FIRST CHECK
invalid signature for subset of users	Signing key rotation removed a key still needed for legacy tokens
Login unavailable during maintenance	Maintenance window overrun — check migration script status

■ CRITICAL / AUTH IMPACT ■ DEGRADATION

3 Diagnostic Steps

- 1

Check **memory utilization trend** over the last 24–48 hours, not just the current snapshot.
- 2

Confirm which `auth-service` version is live in each region — **partial rollbacks** are a recurring cause of regional-only failures.
- 3

If signature validation errors are reported, check whether affected accounts **predate a known key rotation**.

4 Escalation Criteria

- ESCALATE IMMEDIATELY IF:
- Any **login outage affecting more than 1% of active users**.
 - Any incident where users **cannot re-authenticate** after a forced session invalidation.